

Deep Market Study Architecture Blueprint

Candlestick-by-candlestick intelligence, major and inner regression, swing/rest behavior, per-pair lifelong memory, historical similarity, session history, frontend logic, and the independent execution boundary.

Core doctrine: observation is not execution. The study describes the market; the independent permission and packet-validation stack decides whether an entry is allowed.

VERSION	V3 only - no V4
STATUS	Implemented architecture and operating contract
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Engineering blueprint - not financial advice, a performance guarantee, or an instruction to place a trade.

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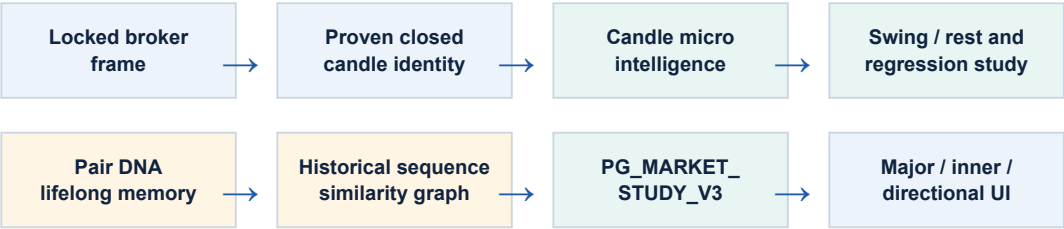
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Architecture at a glance



Execution permission remains a separate, independently validated contract. A BUY or SELL study can never grant entry permission by itself.

- Status: implemented V3 architecture and operating contract
- Updated: 2026-07-29
- Scope: continuous candle intelligence, exact candle memory, behavioral regression, Pair DNA, hierarchical motifs, historical time-to-event evidence, exact normalized paths, adaptive feature governance, concept drift, non-causal cross-pair association, proof certificates, object relationships, retracement confluence, historical similarity, joint path-clock-liquidity timing, and operator presentation
- Version boundary: V3 only. This is not a V4 proposal.

This document is the canonical implementation blueprint for PhoenixGuard's continuous candlestick-by-candlestick V3 study lane. Capture begins with the tracker service, but durable study advancement is automatic and event-driven: every newly proven closed candle can extend the study exactly once, with no manual study baseline and no fixed-length run.

1. What changed, in plain language

PhoenixGuard V3 now keeps two truths separate:

- 1 Market study truth describes what the chart has done, how individual candles behaved, how long swings and rests lasted, what the major and inner trends are, and whether the present sequence resembles prior sequences for the same pair and timeframe.
- 2 Execution truth decides whether an entry is permitted on the current fresh frame.

The study lane may produce a directional BUY study or SELL study while execution remains WAIT. That is intentional. A historical resemblance, regression direction, candle personality, object correlation, or confidence value can inform the operator, but none can grant entry permission.

The implemented lane replaces the idea that every non-permitted situation should be described as only WAIT. The frontend instead presents:

- major trend;
- inner trend;
- current behavior: swing, rest, continuation, or direction change;
- completed-swing 70.5%/71.8% object confluence as explicitly non-executing study evidence;
- historical regression/directional study;
- entry permission as a separate compact safety status.

2. Hard invariants

These invariants apply to every module and payload described below.

Invariant	Required behavior
V3 boundary	All schemas and implementation remain under V3 names and existing V3 routes. No V4 route, schema, or parallel product is created.
Closed-candle causality	Persistent study updates occur only after pair, timeframe, and a unique completed-candle identity are confirmed. Repeated frames for the same close are idempotent.
Forming-candle exclusion	A forming or unproven candle cannot enter the canonical candle intelligence sequence or durable memory.
One authoritative close	Each completed market-study call submits exactly its latest proven close to the exact candle ledger. The durable primary key is (symbol, timeframe, candle identity); an overlapping or restarted observation upserts that row instead of creating another candle.
Exact geometry	Use measured OHLC when available. Otherwise preserve normalized proxy or pixel-proxy coordinates and disclose that coordinate space. Never present pixels as broker prices or pips.
Coordinate isolation	One fingerprint cannot mix PRICE, NORMALIZED_PRICE_PROXY, and PIXEL_PRICE_PROXY candles. Pair DNA partitions behavior aggregates by coordinate space.

Invariant	Required behavior
Retracement coordinate proof	A completed swing and an object's explicit value bounds must use the exact same PRICE, NORMALIZED_PRICE_PROXY, or PIXEL_PRICE_PROXY axis. Normalized screen rectangles are never treated as market values.
Experimental-level honesty	CUSTOM_71_8 is a user-defined, nonstandard experimental level. It is never represented as a classic Fibonacci ratio, a hidden equilibrium fact, or proof of institutional activity.
Study-only output	Every study contract carries study_only: true and execution_authority: false. The market study also carries can_grant_entry_permission: false.
Bounded resources	Candle windows, object lists, caches, stores, graph nodes, and result lists have explicit limits.
Durable publication	JSON stores are validated, finite, size-bounded, locked, written to a same-directory temporary file, fsynced, and atomically replaced. The exact candle ledger separately requires SQLite WAL, synchronous=FULL, and an immediate transaction.
Honest insufficiency	Missing identity, incomplete geometry, too little history, mixed evidence, and persistence failure produce explicit pending/degraded states. They do not invent a trend or match.
Association is not causation	Object/candle outcome statistics are historical associations. Their payload explicitly says they are not causal.
Continuous study	Study advancement is driven by proven closed-candle identity. Repeated frames are idempotent; stopping and restarting capture does not redefine historical study boundaries.
Hierarchical evidence	Motif levels compose single-candle micro-events, 3-5 candle atoms, 7-12 candle compounds, and full swing/rest regimes under deterministic depth and node bounds.
Drift honesty	A drift partition is evidence that a feature distribution changed. It is not a directional prediction or market-cause claim.
Cross-pair honesty	Granger-style variance reduction and mutual information are published only as non-causal historical association. They never establish influence, direction, or permission.
Proof-certificate boundary	A valid study certificate proves canonical derivation integrity against supplied evidence. It does not authenticate the market source, establish causation, or grant entry/execution authority.

3. Implemented end-to-end architecture

MERMAID DIAGRAM / CONTRACT

```

flowchart TD
    A[Locked broker frame] --> B[Candle and chart-object vision]
    B --> C[Pair and timeframe identity checks]
    C --> D[Closed-candle identity resolver]
    D -->|same close| E[Return cached PG_MARKET_STUDY_V3]
    D -->|new proven close| F[Closed visible candle adapter]
    F --> G[Candle Intelligence V3]
    G --> H[Behavioral Sequence V3]
    G --> V[Exact latest close SQLite ledger]
    G --> I[60-field sequence fingerprint]
    H --> I
    B --> J[Bounded object identities]
    J --> I
    G --> W[Bounded candle/object relationship graph]
    J --> W
    G --> X[Completed-swing retracement study]
    J --> X
    X --> W
    X --> M
    I --> K[Same-pair historical search]
    K --> L[Outcome-supported continuation summary]
    G --> AA[Hierarchical motif lattice]
    H --> AB[Historical time-to-event curves]
    G --> AC[Exact normalized historical paths]
    AA --> AD[Shadow feature ontology]
    AD --> AE[Audited promotion or rollback]
    G --> AF[Online distribution drift partitions]
    G --> AI[Restart-safe cross-pair normalized-return coordinator]
    AI --> AG[Exact-timestamp non-causal association graph]
    AA --> AH[Study claim proof certificates]
    AB --> AH
    AC --> AH
    AE --> AH
    AF --> AH
    AG --> AH
    G --> M[Pair DNA cumulative profile]
    H --> M
    J --> M
    N[Major and inner regression slopes] --> O[Directional study fusion]
    L --> O
    G --> O
    M --> P[PG_MARKET_STUDY_V3]
    V --> P
    W --> P
    AA --> P
    AB --> P
    AC --> P
    AE --> P
    AF --> P
    AG --> P
    AH --> P
    O --> P
    P --> Q[tracking_summary and latest_signal]
    Q --> R[Compact live state]
    R --> S[Operator workspace tracking study]
    S --> T[Dashboard trend and regression story]
    U[Independent permission contract] --> T
  
```

The browser renders server truth. It does not write the exact ledger, calculate or persist Pair DNA, build relationship/similarity graphs, mature outcomes, grant permission, or reinterpret a forming candle as closed.

4. Source map

Component	Implementation	Responsibility
Persistence primitives	Backend/src/phoenixguard/study/_persisten ce_v3.py	Cross-thread/process locking, bounded JSON reads, atomic durable writes.
Candle intelligence	Backend/src/phoenixguard/study/candle_int elligence_v3.py	Exact body/wick geometry, taxonomy, personality, prior-candle relation, rejection/acceptance, sequence tokens.
Exact candle ledger	Backend/src/phoenixguard/study/candle_led ger_v3.py	SQLite WAL store with one canonical micro-feature row per stable pair/timeframe/candle identity.
Behavioral sequence	Backend/src/phoenixguard/study/behavioral _sequence_v3.py	Major/inner regression, swing/rest segmentation, durations, path efficiency, transitions, market story.
Pair DNA	Backend/src/phoenixguard/study/pair_dna_v 3.py	Durable pair/timeframe aggregates, bounded recent identities, object/candle outcome associations, and partitioned retracement-confluence outcomes.
Object relationship graph	Backend/src/phoenixguard/study/object_rel ationship_graph_v3.py	Bounded observation graph with explicit candle anchors, co-presence, co-occurrence, proven normalized overlap, and completed-swing retracement/object evidence.
Historical similarity	Backend/src/phoenixguard/study/historical _similarity_v3.py	Explainable fixed-size fingerprints, deterministic similarity, bounded library, graphs, supported outcomes.
Motif, duration, and path research	Backend/src/phoenixguard/study/motif_latt ice_v3.py	Four-level motif composition, Kaplan-Meier-style historical time-to-event curves, and anchor-known normalized historical path reconstruction.
Adaptive feature ontology	Backend/src/phoenixguard/study/adaptive_f eature_ontology_v3.py	Shadow proposals, versioned temporal/leakage promotion gates, promoted-only public snapshots, and reversible audited rollback.
Concept drift	Backend/src/phoenixguard/study/concept_dr ift_v3.py	Adjacent-window KS evidence and deterministic pair/timeframe regime partition identities.
Cross-pair association	Backend/src/phoenixguard/study/cross_pair _association_v3.py	Exact shared-timestamp, compatible-coordinate Granger-style proxy and mutual-information graph, always explicitly non-causal.
Cross-pair coordinator	Backend/src/phoenixguard/study/cross_pair _coordinator_v3.py	Restart-safe bounded normalized-return state, exact timestamp synchronization, atomic update, and insufficient-peer/support abstention without fabricated edges.
Study claim proofs	Backend/src/phoenixguard/study/study_clai m_proof_v3.py	Deterministic content-addressed integrity certificates over closed-candle evidence, inputs, order domain, coordinate space, and derivation.
Live study coordinator	Backend/src/phoenixguard/study/market_stu dy_service_v3.py	One idempotent study per proven close, prior-outcome maturation, deterministic research rebuild, cross-pair coordination, claim-proof attachment, compact public study, and directional read.
Tracker integration	Backend/src/phoenixguard/mobile_api/windo w_tracker.py	Identity gates, close adaptation, regression context, object reduction, study invocation, safe degradation.
Live-state projection	Backend/src/phoenixguard/mobile_api/live_ state_v3.py	Preserves bounded market_study_v3 in tracking and latest-signal compaction.
API allowlist projection	Backend/src/phoenixguard/mobile_api/app.p y	Projects explicitly selected nested study evidence instead of relying on a generic depth walk.
Operator projection	Backend/src/phoenixguard/mobile_api/opera tor_workspace_v1.py	Keeps the fixed top-level schema and projects the study inside tracking; history remains evidence-only.
Dashboard	Frontend/dashboard/static/window_tracker_ dashboard.html	Major/inner/regression story, continuous candle-by-candle history, exhaustive label mode, separate permission.

5. Closed-candle causality and idempotency

5.1 Authority chain

The causal key is not a frame number. It is the latest completed-candle event resolved for one confirmed (pair, timeframe) context.

- 1 `resolve_closed_candle_identity_v3(...)` produces a `closed_candle_key`, monotonic `closed_candle_sequence`, and `PG_CLOSED_CANDLE_IDENTITY_STATE_V3` state.
- 2 The tracker requires confirmed market and timeframe identities and a non-empty close key.
- 3 Visible rows are filtered by their explicit closed/forming fields. When a detector does not label closure, the tracker treats the right-most candle as forming and only earlier visible rows as historical closed candles.
- 4 `adapt_tracker_candle_v3(...)` requires `proven_closed: true`, a non-empty event key, and matching candle identity when both the proof and row identify the candle. It validates geometry before it adds `is_closed: true`.
- 5 `analyze_candle_sequence_v3(..., require_closed=True)` rejects every row that does not contain an explicit closed marker. Thus the analysis module itself never infers closure from geometry or list position.
- 6 The market-study cache key is (symbol, timeframe, closed_candle_key). Repeated screenshots of the same forming candle return the prior result without another persistence update.

The tracker requires at least five visible candles so that, under the conservative right-edge policy, at least four proven historical candles remain for the first sequence fingerprint.

5.2 Resolver-to-study identity bridge

Tracker `track_id`, generic `id`, and rolling sequence index values are positional display identifiers. They restart or shift when the visible window is reacquired and therefore are never stable candle identity, chronological order, or persistence proof.

For a screenshot-only candle that has no stable source timestamp, the tracker may promote identity into the study lane only when all of the following are present and consistent:

- the proof source is exactly `PG_CLOSED_CANDLE_IDENTITY_STATE_V3`;
- the resolver supplies a non-empty stable closed-candle event key;
- the resolver supplies a non-negative monotonic event sequence for the same pair/timeframe;
- the row appears in the resolver's current `stable_visible_candle_bindings` contract; and
- a prior event appears there only after the resolver uniquely re-observes it at the exact predecessor position implied by the confirmed transition count on the current candle axis.

The unique prior match must clear the resolver's similarity and separation checks and form a contiguous X-axis chain into its successor. An arbitrary shifted window, a detector reacquisition, or a merely similar positional candle cannot manufacture this proof. The bridge adds `identity_stable`, `stable_candle_identity`, `identity_proof_source`, and `closed_candle_sequence` only to rows covered by that proof. All other positional rows remain valid for bounded visual analysis but cannot enter lifelong candle counting or mature an outcome. The binding list is sorted by event sequence and capped at 32 current-frame rows. Duplicate indexes, keys, sequences, malformed proof metrics, a stale pair/timeframe context, an unsupported proof source, or an omitted/ambiguous/off-screen row fails closed; the tracker never backfills it from position. A pair or timeframe change resets the resolver history to one latest-close baseline at sequence zero.

A changed immutable source ID is not automatically one candle. A source forming-bar identity that becomes the current closed bar proves one transition. Numeric source time identities may also prove one transition only when their delta equals exactly one declared timeframe after seconds, milliseconds, microseconds, or nanoseconds normalization. A larger/backward/malformed timestamp or an arbitrary changed bar ID remains `SOURCE_BAR_GAP_UNPROVEN` unless the visible reacquisition chain enumerates every intervening close. Its event sequence does not advance, so it cannot mature an $N \rightarrow N+1$ outcome.

5.3 What can and cannot be persisted

- The current forming candle may remain visible on the chart, but is excluded from the canonical closed sequence and every durable study mutation.

- Detector refinement within the same closed-candle event may improve display geometry, but the market-study event remains keyed to the same causal close.
- A pair change, timeframe change, identity rebind, or missing close key produces a pending study.
- A malformed historical candle is skipped by the tracker. If fewer than four valid closed candles remain, the whole market study reports `INSUFFICIENT_HISTORY`.

5.4 Exact candle ledger

Schema: `PG_CANDLE_LEDGER_V3`

Default file: `candle_ledger_v3.sqlite3`

The ledger is the authoritative distinct-candle store. After a study has at least four valid closed candles, the service takes only the newest studied candle, marks the proven `closed_candle_key` as its stable identity, and submits that one row. The SQLite primary key is (`symbol`, `timeframe`, `candle_identity`). The result is therefore one authoritative close per study event, even though the analyzer used a rolling window of up to 128 candles.

An insert retains the exact allowlisted source values, canonical OHLC/proxy geometry, full body/wick measurements and ratios, taxonomy, personality, regime, relation, interaction evidence, and sequence position. An upsert refreshes the canonical evidence and increments `observation_count`; it does not increment `unique_candle_count`. Raw screenshots, arbitrary detector payloads, model inputs, orders, and execution authority are never stored.

The write contract is intentionally strict:

- the candle must be `STUDIED`, explicitly closed, study-only, and non-executing;
- identity must be explicit and stable; synthetic positional names are rejected as identities;
- coordinate space and its required source fields must agree;
- one batch is validated completely before mutation, and conflicting duplicate identities fail the whole batch;
- service integration submits one row; the reusable ledger accepts at most 4,096 rows per atomic batch;
- default capacity is 1,000,000 unique rows, configurable only within 1 to 10,000,000; a capacity breach rolls back the entire batch;
- recent reads are bounded to 512 rows.

Every connection requires WAL mode, synchronous=FULL, foreign keys, a default 5,000 ms busy timeout, and `BEGIN IMMEDIATE` for initialization and writes. A schema mismatch or a pre-existing table without V3 schema authority requires an explicit migration; the store never silently adopts or resets it.

6. Candle Intelligence V3

Schema: `PG_CANDLE_INTELLIGENCE_V3`

6.1 Supported coordinate sources

The geometry adapter selects exactly one representation, in this order:

- 1 broker/market open, high, low, close -> `PRICE`;
- 2 open_proxy, high_proxy, low_proxy, close_proxy -> `NORMALIZED_PRICE_PROXY`;
- 3 measured wick/body pixel coordinates -> `PIXEL_PRICE_PROXY`.

For pixel geometry, image Y increases downward. The module negates Y to create a local price-like axis, but retains the original pixel values under `source_values`. If only body top/bottom is known, the measured candle direction determines which body edge is open and which is close. Unknown direction is rejected rather than guessed.

Every candle envelope is validated:

- all numeric fields must be finite;
- `high` $\geq$ `low`;
- `high` cannot fall below the body;

- low cannot rise above the body;
- total range must be positive;
- pixel body top cannot be below pixel body bottom.

6.2 Exact micro features

For a valid range $R = \text{high} - \text{low}$, the record preserves:

- body size $\text{abs}(\text{close} - \text{open})$;
- upper wick $\text{high} - \max(\text{open}, \text{close})$;
- lower wick $\min(\text{open}, \text{close}) - \text{low}$;
- body/range, upper-wick/range, lower-wick/range, and total-wick/range;
- body/total-wick and each wick/body when denominators are non-zero;
- close location within the full range;
- range relative to the sequence median;
- exact source values and canonical OHLC/proxy values.

These are measured values. A pixel-space wick is not translated into pips until a separate, validated price-scale mapping exists.

6.3 Implemented taxonomy

The deterministic candle taxonomy includes:

- doji and long-legged doji;
- spinning top;
- bullish/bearish marubozu;
- upper- and lower-wick rejection;
- bullish/bearish impulse;
- bullish/bearish balanced candle;
- balanced indecision.

The personality layer adds context such as:

- liquidity rejection high/low;
- breakout acceptance up/down;
- higher/lower-price rejection;
- expansion up/down;
- compression;
- assertive or controlled buying/selling;
- indecision.

Relative behavior records inside/outside bars, higher-high/higher-low and lower-high/lower-low relationships, mixed extensions, and overlapping ranges. Rejection checks whether the wick swept the prior high/low and closed back inside. Acceptance requires a close beyond the prior extreme with a body ratio of at least 0.40.

Each candle receives a stable 16-character token derived from direction, type, personality, relation, and regime. The full sequence signature is the SHA-256 digest of the ordered candle tokens. The service bounds one live sequence to 128 candles; the reusable analyzer supports 512 by default and validates caller limits up to 4,096.

7. Behavioral Sequence V3

Schema: PG_BEHAVIORAL_SEQUENCE_V3

This module describes observed behavior instead of forcing every non-entry state into WAIT.

7.1 Per-candle state

Each closed candle becomes one of:

- UP_SWING;
- REST;
- DOWN_SWING.

The state uses close-to-close movement normalized by the sequence median range. A candle is treated as resting when normalized movement is at most 0.30 and its body, range multiple, or indecision type also shows compression. This is a descriptive state, not an entry signal.

7.2 Segments and duration

Consecutive equal states form a segment. Every segment records:

- start/end indices and candle IDs;
- candle count and duration in seconds;
- direction and price change;
- absolute change in median-range units;
- high, low, and path efficiency;
- average body, upper wick, lower wick, and range multiple;
- prior and next state.

Duration uses actual timestamps when they parse and are monotonic. Otherwise it falls back to `candle_count * timeframe_seconds`. Swing summaries are split into up and down. Rest summaries also count later up breakouts, down breakouts, and unresolved rests.

7.3 Major and inner trends

Major and inner trend are always separate regression scopes:

- major uses the full bounded studied sequence;
- inner uses the newest eight candles, or the available shorter sequence with a minimum of two.

Each trend reports normalized slope, directional efficiency, net change, strength, and one of UP, SIDEWAYS, DOWN, or INSUFFICIENT_HISTORY. A directional label requires normalized slope magnitude of at least 0.08 and directional efficiency of at least 0.22.

The current compact story is intentionally simple: major trend, inner trend, and the current swing or rest duration.

7.4 Transition memory

The module builds observed counts and conditional probabilities for all transitions among UP_SWING, REST, and DOWN_SWING, both candle-by-candle and segment-by-segment. Pair DNA later accumulates these transition counts by coordinate space.

8. Pair DNA: durable per-pair behavioral memory

Schema: PG_PAIR_DNA_STORE_V3

Default file: pair_dna_v3.json

One profile is keyed by a SHA-256-derived identifier for canonical (`symbol`, `timeframe`). This prevents CAD/JPY M5 behavior from being silently mixed with CAD/JPY M1 or another pair.

8.1 Persistent profile fields

Group	Persisted evidence
Identity	Pair ID, symbol, timeframe, first/last observation, update ordinal.
Candle population	Observation and candle counts; direction, candle-type, and personality counts.
Wick/body behavior	Body and wick ratio sums; rejection, acceptance, upper-sweep, and lower-sweep counts.
Behavioral population	State counts, segment counts, candle/duration/change sums, transition counts.
Trend/regime	Major trend, inner trend, and regime counts.
Source truth	Coordinate-space counts and coordinate-partitioned behavioral metrics.
Objects	Counts of bounded detected object types.
Outcomes	Marginal and pairwise feature association support, direction counts, success counts, and realized-return sums.
Retracement confluence	Completed-study counts and bounded empirical buckets partitioned by current observation regime, swing side, coordinate space, level, and object type.
Bounded recency	Recent sequence summaries and recently seen sequence IDs.
Exact incremental boundary	One locked candle order domain, its monotonic high-water mark, completed-segment boundary high-water mark, current open segment, accepted counts, and audited skip/conflict counts.
Whole-sequence replay protection	Bounded exact recent ring plus segmented SHA-256 Bloom state.

Derived reads calculate average wick/body ratios, rejection/acceptance/sweep rates, transition probabilities, segment averages, smoothed outcome direction probabilities, success rates, and average realized returns. These derived fields are not written back as independent authority.

8.2 Object and candle association features

Implemented association tokens include the latest candle type, personality, behavior state, regime, individual object types, candle/object pairs, personality/object pairs, and pairs of object types. The contract calls them `MARGINAL_AND_PAIRWISE_FEATURE_ASSOCIATION` and explicitly sets `causal: false`.

At most 768 sorted feature tokens can be proposed by one matured study. Object pair construction is bounded to the first 32 distinct sorted object types. A profile stores at most 2,048 outcome association keys; new keys beyond that limit increment `association_overflow_count` instead of evicting established evidence. The object-type and regime maps are separately bounded to 256 and 128 keys, respectively, with bounded overflow accounting. The compact live result publishes the 12 strongest associations by support.

8.3 Bounded object relationship graph

Schema: `PG_OBJECT_RELATIONSHIP_GRAPH_V3`

The current-study graph records only what one observation scope proves:

- `ANCHORED_TO_CANDLE` is a directed edge and requires an explicit object candle identity that exactly matches a studied candle;
- `OBSERVED_WITH` links an unanchored object to the latest studied candle because both were present in the same study call, but its proof explicitly says `anchor_inferred: false`;
- `CO_OCCURS` means two objects were supplied in the same observation scope;
- `OVERLAPS` exists only when both objects have valid normalized rectangles with positive intersection; the edge reports intersection area and intersection-over-union.

The tracker boundary preserves every distinct source object key, including real support/demand and resistance/supply rows that previously collapsed under an empty generic identity. When the exact chart-image width and height are available, its pixel bbox, touch points, or anchor-wick points are converted once into bounded normalized evidence and the raw pixel coordinates are discarded. Without that explicit frame size, pixel, point-only, or malformed geometry is never guessed into a normalized overlap. An object with no source identity receives an observation-local ordinal and is explicitly marked `OBSERVATION_ONLY`,

never stable. Objects retain bounded identity scope, direction, confidence, lifecycle, geometry, explicit/matched/unresolved candle associations, and an evidence digest. Every edge is observational and non-causal.

The reusable graph defaults to 64 object nodes, 16 candle nodes, 512 edges, and 32 points per object. Its absolute validated limits are 256 object nodes, 64 candle nodes, 4,096 edges, 128 points per object, and 4,096 input rows of either kind. The live service deliberately uses tighter caps: 32 objects, eight candle nodes, 128 edges, and eight points per object after the tracker has already reduced its object evidence to at most 64 rows. Omitted objects, candle anchors, and edges are counted, and the graph reports READY_TRUNCATED rather than pretending it is complete.

8.3.1 Completed-swing retracement and object confluence

Schema: PG_RETRACEMENT_CONFLUENCE_STUDY_V3

This is a study inside the V3 object relationship graph, not a forecast lane and not another route. It evaluates two deliberately distinct ratios:

Level	Classification	Contract
OTE_70_5 / 0.705	ICT-style OTE reference	A non-classic community reference studied alongside PhoenixGuard's disclosed 61.8%-78.6% band. It is not labeled a standard Fibonacci ratio or the exact midpoint of those two boundaries.
CUSTOM_71_8 / 0.718	User-defined experimental, nonstandard	Preserved exactly so PhoenixGuard can measure the user's hypothesis. It carries experimental: true, user_defined: true, and standard_fibonacci: false.

The catalog also discloses 61.8% and 78.6% as reference-band boundaries; the current confluence study does not silently substitute either boundary for 70.5% or 71.8%. In particular, 71.8% is not half an impulse move (0.5) and it is not half of 1.618 (0.809). PhoenixGuard therefore learns its observed association instead of embedding an unsupported success claim.

A swing is eligible only when all included candles are explicitly closed, their identities and sequence order are proven, their value coordinate space is uniform, and a later closed candle confirms the ending pivot. A latest/forming candle cannot confirm its own pivot. Ambiguous outside candles that are simultaneously local highs and lows are excluded because their intrabar order is unknown. The resulting swing carries stable start, end, and confirmation candle identities plus the exact completion proof.

For swing range $R = \text{high} - \text{low}$, the evaluated value is:

- upward swing: $\text{end_value} - \text{ratio} * R$;
- downward swing: $\text{end_value} + \text{ratio} * R$.

An object participates only when it is an identity-stable Order Block, Fair Value Gap/price imbalance, or crowded/consolidation area with explicit one-dimensional value_bounds. Its value_coordinate_space must exactly match the swing. For screenshot geometry the tracker maps the original pixel box from [top, bottom] to [-bottom, -top], the same disclosed PIXEL_PRICE_PROXY axis used by candle intelligence. It still retains normalized rectangles only for screen overlap; it never treats normalized screen Y as broker price.

The graph records RETRACEMENT_LEVEL_OVERLAPS_OBJECT when the value lies inside the object bounds, or RETRACEMENT_LEVEL_NEAR_TOUCHES_OBJECT when the nearest boundary is within the configured fraction of the completed swing range. Every completed observation preserves the level, swing, object family, coordinate space, distance, tolerance, regime, side, and stable evidence digest. It also states causal: false, study_only: true, observation_only: true, and execution_authority: false.

Pair DNA accepts only those completed, stable observations after the existing one-candle outcome matures. It partitions them by exact (pair, timeframe, observation regime, swing side, coordinate space, level, object type) so real-price observations never mix with screenshot proxies and 70.5% never mixes with 71.8%. The regime dimension is explicitly CURRENT_STUDY_FRAME_AT_CONFLUENCE_OBSERVATION: it describes the regime when PhoenixGuard recorded the object-confluence observation, not a reconstructed claim about the older swing's event-time regime. Derived reads therefore expose observation_regime and regime_basis rather than silently presenting today's frame classification as historical swing truth. They always publish support beside empirical direction frequencies, labeled next-candle alignment rates, and average side-adjusted return. A bullish completed swing treats an UP outcome as aligned; a bearish completed swing treats DOWN as aligned, and bearish signed returns are multiplied by -1 before aggregation. The unrelated overall directional-study success label is never presented as retracement success. These are historical frequencies, not calibrated forecast

probabilities. A separate segmented Bloom identity prevents the same confluence from being counted again through overlapping windows or service restarts.

8.4 Exact incremental counting, replay protection, and bounds

Pair DNA has three independent idempotency layers:

- 1 Whole-sequence replay protection. The newest 512 exact sequence IDs remain in a bounded ring. All accepted sequence IDs also enter a SHA-256 segmented Bloom design: 20 sealed segments, 32,768 bits and 16 hashes per segment, and 512 insertions per segment. Total design capacity is 10,240 sequence IDs per pair; the union false-positive probability across all full segments is below 1e-9. At 10,240 identities it fails closed and requires sharding instead of opening an unsafe segment or forgetting old IDs. A safe legacy 16,384-bit/five-hash filter can be carried as a separately validated migration attachment.
- 2 Candle and segment boundary deduplication. Each pair/timeframe profile locks to exactly one chronological domain: CLOSED_TIMESTAMP_V1 for a stable source close timestamp, or TRACKER_EVENT_SEQUENCE_V3 for a resolver-proven event key plus monotonic event sequence. Domains are never numerically compared or mixed. Once locked, a row from the other domain is audited under skipped_order_domain_conflicts and cannot change lifelong aggregates. Within the selected domain, the marker must move strictly past the per-pair candle high-water mark. Overlapping rows, out-of-order backfills, missing/non-monotonic order, and unstable positional identities are audited but not counted. A segment is counted only after the following segment proves its end boundary and both start/end markers resolve in that same locked domain. The open segment and completed-boundary high-water mark prevent the same duration, transition, or swing/ rest segment from being counted again in the next rolling window.
- 3 Retracement-study replay protection. Completed confluences have stable study IDs derived from the proven swing, exact level, and stable object. A separate segmented SHA-256 Bloom plus bounded recent-ID ring prevents an overlapping window or restart from incrementing the same empirical bucket twice. Capacity fails closed instead of forgetting old confluence identities.

observation_count counts accepted unique study envelopes. candle_count and candle distributions count only causally new stable closes selected by the monotonic boundary ledger, never repeated rows from overlapping analysis windows. The separate SQLite unique_candle_count remains the authoritative exact row count, while Pair DNA is the cumulative behavioral aggregate. A legacy profile with historical aggregates but no boundary ledger establishes a conservative baseline without replaying its visible window.

The remaining profile bounds are:

- default maximum profiles: 128, configurable up to 4,096;
- exact recent sequence IDs/summaries: 512 by default, configurable up to 4,096;
- a confirmed exact duplicate returns DUPLICATE_IGNORED;
- a Bloom-positive ID outside the recent exact ring returns POSSIBLE_DUPLICATE_IGNORED and leaves counts unchanged;
- profile capacity never silently evicts a pair's lifelong aggregates; it fails closed and asks for sharding or an explicit limit increase.

9. Historical fingerprint and similarity engine

Fingerprint schema: PG_SEQUENCE_FINGERPRINT_V3

Store schema: PG_HISTORICAL_SEQUENCE_STORE_V3

Default file: historical_sequences_v3.json

9.1 Explainable 60-value fingerprint

The newest maximum 64 closed candles are resampled into a fixed representation:

Component	Dimensions	Normalization
Price path shape	16	Change from first close / median range, clipped to +/-8 ranges, scaled to +/-1.
Close-delta shape	8	Close-to-close delta / median range, clipped to +/-4 ranges, scaled to +/-1.

Component	Dimensions	Normalization
Body shape	8	Body/range in $[\emptyset, 1]$.
Upper-wick shape	8	Upper wick/range in $[\emptyset, 1]$.
Lower-wick shape	8	Lower wick/range in $[\emptyset, 1]$.
Direction distribution	3	Bullish, neutral, bearish proportions.
Behavior distribution	3	Up swing, rest, down swing proportions.
Major trend	3	One-hot up, sideways, down.
Inner trend	3	One-hot up, sideways, down.
Total	60	Fixed and validated.

The fingerprint also retains the newest 32 ordered candle tokens, unique object types, regime, coordinate space, latest candle summary, symbol/timeframe, and optional matured outcome. A SHA-256 digest covers the immutable fingerprint core; reads reject a digest mismatch.

9.2 Similarity function

The current deterministic score is transparent and fixed:

Evidence	Weight
Normalized price path	0.28
Candle delta path	0.10
Body and both wick shapes	0.22
Direction and behavior distributions	0.12
Major and inner trend match	0.08
Ordered/set candle-token similarity	0.10
Detected-object Jaccard similarity	0.05
Regime agreement	0.05

The live search is same-pair and same-timeframe by default, requests the best eight matches, and requires score ≥ 0.55 . The query cannot match itself. Results explain which components aligned and name shared object types.

The optional historical graph view is distinct from the current-study object graph. It is an undirected sequence-similarity graph over at most the newest 256 fingerprints. The reusable default is eight edges per node and the validated caller range is one to 32. The live service requests six edges per node at similarity ≥ 0.65 , refreshes the cached graph on first use and every 12th close, then publishes at most 64 same-pair nodes and 128 edges in its compact study. The public app allowlist reduces that again to graph metadata and at most 24 scalar-only edges. This is an inspection surface, not part of trade permission.

9.3 Bounded storage

- Default global entries: 2,048.
- Default entries per pair/timeframe: 512.
- Adding an identical fingerprint updates its outcome while preserving insertion ordinal.
- Pair overflow removes the oldest entries for only that pair/timeframe.
- Global overflow removes the oldest entries globally.
- Store validation rejects duplicates, invalid dimensions, non-finite values, schema mismatch, and configured-bound overflow.

Pair DNA preserves cumulative aggregates, while the historical store deliberately preserves a bounded set of inspectable sequence examples. They have different retention contracts.

10. Outcome maturation and historical continuation

Outcome labels are causal and delayed by one proven candle close.

For pair/timeframe close N:

- 1 Build and search the fingerprint for sequence N without using an unseen future outcome.
- 2 Load the bounded durable pending record for N-1 and require the current resolver event sequence to equal the pending event sequence plus exactly one. A replay, gap, or out-of-order event reports `SKIPPED_UNPROVEN_ONE_STEP_HORIZON` and cannot mature the pending record.
- 3 Find the prior latest candle again among the earlier candles of the current frame, using its stable source timestamp or its resolver-proven stable event identity. Positional tracker IDs do not qualify.
- 4 Require that uniquely re-observed candle and the newest candle share the current study's coordinate space. Compare their closes on that one current-frame axis and normalize the move by the current sequence median range.
- 5 Label it UP above $+0.04$, DOWN below -0.04 , otherwise REST.
- 6 Set horizon to one candle and record whether the prior BUY/SELL study matched the actual side.
- 7 Enrich the prior historical fingerprint and merge the prior completed study into Pair DNA with that matured outcome, including any stable completed retracement/object observations from that prior study.
- 8 Add the current fingerprint as unlabeled, ready to mature at N+1.

If the prior close is not re-observed on the current frame, its identity is ambiguous, or coordinate spaces differ, maturation reports `SKIPPED_UNPROVEN_COORDINATE_CONTINUITY`. It does not compare raw pixel or normalized-proxy values from separately scaled screenshots, label the historical fingerprint, or update outcome associations. Successful outcomes carry `coordinate_continuity: CURRENT_FRAME_REOBSERVATION`. This ordering prevents both current-search leakage and chart-autoscale/pan leakage.

The pending record retains the resolver's stable identity, proof source, and event sequence needed for that exact +1 check. A newly reacquired or horizontally shifted pixel window without the resolver's unique predecessor proof cannot satisfy the horizon indirectly, even if its candle shapes resemble the pending sequence.

Historical continuation is published only when at least three selected matches have a labeled outcome. Similarity-weighted UP/DOWN/REST probabilities use a small symmetric prior. If the top two probabilities differ by less than 0.08 , the result is `MIXED_EVIDENCE` rather than a forced side. Confidence combines dominant probability, mean similarity, and a bounded support factor.

The Pair DNA association view uses add-one smoothing. The historical correlation report includes support, lift versus the pair baseline, and a 95% Wilson interval for the dominant outcome. Both are descriptive association reports, never causal claims.

The causal pending record is persisted atomically in `pending_outcomes_v3.json`, bounded to 64 pair/timeframe entries and protected by the same cross-process lock discipline as the other JSON study stores. Each entry retains the fingerprint plus only the evidence needed for one-step maturation: at most 16 compact candles, 16 behavior states, 16 segments, and 16 objects, together with the prior study side and latest candle identity/coordinate space. When a 65th pair is written, the oldest pending ordinal is evicted; Pair DNA and the exact candle ledger are not evicted. A restart between N and N+1 can therefore reload the bounded prior study and mature it once the next proven, same-axis close arrives.

11. Live Market Study service

Schema: `PG_MARKET_STUDY_V3`

MarketStudyServiceV3 owns:

- `PairDNAStoreV3`;
- `CandleLedgerStoreV3`;
- `HistoricalSequenceStoreV3`;
- an atomic bounded pending-outcome journal plus a process-local hot cache;
- a bounded current-study object relationship graph and cached historical similarity graph;
- an idempotent result cache keyed by pair, timeframe, and closed-candle key, capped at 64 entries.

The public result is deliberately compact:

TEXT DIAGRAM / CONTRACT

```

PG_MARKET_STUDY_V3
|- identity: symbol, timeframe, closed key/sequence, sequence ID, observed time
|- regression: major trend, inner trend, current pressure, regime
|- candle_intelligence
|   |- counts, baseline, sequence signature
|   |- latest candle
|   `-- newest 12 exact candle studies
|- candle_ledger
|   `-- this-close insert/update and exact pair unique/observation counts
|- behavior
|   |- major and inner trend
|   |- current state and current segment
|   |- swing/rest summaries and transition matrix
|   `-- plain-language market story
|- pair_dna
|   |- cumulative counts and rates
|   `-- strongest 12 outcome associations
|- object_relationship_graph
|   |- bounded candle and market-object nodes
|   `-- proven anchor, observed-with, overlap, and co-occurrence edges
|- historical_similarity
|   |- strongest eight matches
|   |- support-gated continuation
|   `-- bounded historical sequence graph
|- outcome_maturation
|   `-- matured, skipped-coordinate-continuity, or no-prior status
`-- directional_read: BUY, SELL, or HOLD study

```

11.1 Directional study fusion

The directional read fuses independent descriptive votes:

- major regression: weight 0.52;
- inner regression: weight 0.30;
- a support-qualified historical continuation: weight 0.18.

When no directional vote exists, the latest completed candle may provide only a very weak fallback. Opposing votes reduce confidence. Reasons expose every included side and evidence confidence. The result is marked `DIRECTIONAL_STUDY` or `INSUFFICIENT_EVIDENCE` and always includes `can_grant_entry_permission: false`.

This is the operator's one studied direction, not a second execution lane and not two simultaneous forecast routes.

12. Tracker, live state, operator, continuous history, and frontend flow

12.1 Tracker integration

PhoenixGuardWindowTrackingAdapter creates the study after visible candles, global/local/current regression slopes, major trend context, consolidation, structure boxes, historical structure, and support/resistance zones have been calculated.

The tracker reduces chart objects to at most 64 rows, including production Smart Money Context Order Blocks and Fair Value Gaps plus qualified crowded/liquidity-pool zones. Every row starts with object type, bounded ID, direction, and confidence. Real source key, zone_id, object_id, and equivalent keys remain distinct display identities instead of collapsing under role/type, but the presence of a key does not itself prove lifelong stability. An object becomes identity-stable only when all of its source candle indexes resolve through the current stable-binding contract, or when an upstream source explicitly supplies a non-positional stable-identity proof. The tracker hashes the object type, direction, and stable candle anchors into the durable object identity. With the exact captured image width and height, pixel bbox, touch points, and anchor-wick points are normalized once into bounded [0, 1] evidence and raw pixel geometry is stripped. Without those dimensions the tracker does not guess a normalization. A valid pixel box separately publishes bounded value_bounds on the negated-Y PIXEL_PRICE_PROXY axis so it can be compared only with candles measured on that same captured chart axis. Anonymous objects receive an observation-local ordinal with identity_scope: OBSERVATION_ONLY; keyed-but-unproven objects receive the same scope and are never advertised as stable across frames. When actually present, rows also preserve lifecycle/first/last-seen/duration evidence. Stable candle associations are published only from the verified resolver anchors. All graph edges are study-only observations, never causal or executable. The tracker derives regime as sideways, uptrend, downtrend, or transition. The completed study is placed in both:

- tracking_summary.market_study_v3;
- latest_signal.market_study_v3.

Any exception in the study lane is caught, logged, and converted to a DEGRADED study. It must not interrupt capture, permission safety, or atomic session publication.

12.2 Live-state and operator projection

The compact live-state allowlists preserve market_study_v3 in both tracking and latest signal. The app does not apply the old generic recursive depth limiter to this evidence tree. Its dedicated allowlist first requires study_only: true and execution_authority: false, then selects only bounded identity/regression fields, candle summary and latest ratios/interactions, behavior, support-gated historical matches/continuation, bounded similarity edges, Pair DNA counts and 12 associations, the directional read, and a compact retracement summary. That summary exposes only the two fixed level definitions, current graph support, full Pair DNA per-level support, and at most 16 empirical partitions; it strips swing/object IDs, value bounds, raw buckets, and dedupe state. Raw OHLC/source pixel geometry, full fingerprints, model inputs, persistence metadata, and arbitrary nested keys do not cross the public app boundary.

The operator workspace keeps its established fixed top-level schema and publishes the live study under:

TEXT DIAGRAM / CONTRACT
operator_workspace.tracking.market_study_v3

Do not introduce a V4 route or an extra unversioned top-level market-study key. The frontend may read the tracking study first and use a latest-signal copy only as a compatibility fallback.

12.3 Continuous closed-candle history

History rows are not a list of permission decisions and do not depend on an operator-created study run. A completed-candle event receives a compact market-study snapshot only when both its closed_candle_key and monotonic closed_candle_sequence exactly match the study. Reacquired batches may carry a distinct study with each confirmed event. The newest live study is never copied backward onto older events; an unproven sequence gap remains an UNKNOWN_GAP without invented candle behavior. Each correctly attributed completed-candle observation can carry or derive:

- actual candle direction;
- major trend;
- inner trend;
- regression/directional study side;
- behavior classification: movement, rest, continuation, or direction change;
- historical-match support or difference when comparable prior evidence exists;
- observed time and monotonic closed-candle sequence.

Retained summaries remain neutral about execution. A study with no proven completed candles is described as insufficient regression history, not as a permanent WAIT trend.

12.4 Frontend contract

The V3 dashboard is a chart-led study workspace:

- the forecast-control panel, Forecast navigation, dual-path forecast presentation, and public two_candle, scene_forecaster, lstm, and prediction overlay families are retired from the V3 workspace;
- retired forecast families are absent from ALL_OVERLAY_FAMILIES, presets, public view routing, removed forecast controls, and diagnostic rendering, so Show all cannot re-enable them;
- one concise market story shows major trend, inner trend, and historical regression;
- current movement/rest/continuation is described without turning it into permission;
- entry permission remains a separate compact block;
- history is titled and rendered as a continuous candle-by-candle regression study;
- Show all plus Labels on activates labels-show-all: collision solving is bypassed and both label-collision-hidden and label-policy-hidden labels are forced visible even when they overlap or cluster;
- hover and reduced-label modes remain available when the operator wants less clutter.

The dashboard may derive a readable fallback from history and overlays during warm-up, but server study fields take precedence once PG_MARKET_STUDY_V3 is available.

13. Storage and persistence contract

13.1 Location

The study root resolves in this order:

- 1 explicit constructor path;
- 2 PHOENIXGUARD_MARKET_STUDY_DIR;
- 3 for canonical live runtime roots, data/mobile_api/window_tracker/market_study_v3 under the project;
- 4 for isolated/test roots, a sibling market_study_v3 directory.

This deliberately keeps lifelong study memory outside launcher-cleaned runtime/live output.

13.2 Files

TEXT DIAGRAM / CONTRACT

```
market_study_v3/
|- pair_dna_v3.json
|- pair_dna_v3.json.lock
|- candle_ledger_v3.sqlite3
|- candle_ledger_v3.sqlite3-wal      # SQLite-managed while active
|- candle_ledger_v3.sqlite3-shm      # SQLite-managed while active
|- historical_sequences_v3.json
|- historical_sequences_v3.json.lock
|- adaptive_feature_ontology_v3.json
|- adaptive_feature_ontology_v3.json.lock
|- cross_pair_coordinator_v3.json
|- cross_pair_coordinator_v3.json.lock
|- pending_outcomes_v3.json
|- pending_outcomes_v3.json.lock
```

Each JSON root repeats its schema version and the study/execution boundary. A JSON document larger than 16 MiB fails closed, and encoding rejects NaN and infinity. The SQLite ledger carries both the V3 ledger schema and SQL schema version in ledger_meta; its default distinct-row capacity is 1,000,000. WAL and shared-memory sidecars are SQLite implementation files, not separate sources of truth and not files for manual editing.

Motif, survival, path, and proof-certificate results rebuild deterministically from retained continuous history. The pair-scoped adaptive ontology owns a versioned, bounded audit store. Drift detector snapshots and append-stable regime partitions are persisted inside Pair DNA without raw feature windows. The cross-pair coordinator owns the versioned, atomically replaced, bounded cross_pair_coordinator_v3.json normalized-return store so pair updates arriving at different times can be matched only on exact shared closed timestamps. None of these stores may retain unbounded feature windows, raw price geometry, or raw cross-pair evidence.

13.3 Concurrency and durability

- For JSON stores, a path-specific re-entrant mutex serializes threads in one process.
- A JSON sidecar byte lock serializes processes: msvcrt on Windows, fcntl on POSIX.
- JSON lock acquisition has a five-second default timeout and checks every 20 ms.
- The JSON writer encodes and validates the complete replacement document first, writes a same-directory temporary file, flushes, calls fsync, and publishes with os.replace.
- A failed JSON write cleans up its temporary file and leaves the prior document authoritative.
- The candle ledger uses a process-local re-entrant mutex plus SQLite WAL and BEGIN IMMEDIATE for cross-process serialization. It validates the whole proposed batch and capacity before applying any insert or update, then commits once or rolls back completely.

No automatic repair or silent reset is attempted for malformed JSON or an incompatible SQLite schema. Operators must preserve the bad store for diagnosis, restore a known-good copy, or run a deliberate, versioned rebuild/migration.

14. Failure and degraded-state matrix

Condition	Public study state	System behavior
Pair/timeframe not confirmed	PENDING	No fingerprint or persistence mutation.
Completed-candle key absent	PENDING	Wait for causal identity; repeated frames do not fabricate an event.
Study root not configured	UNAVAILABLE	Tracker and permission continue; no lifelong study write.
Fewer than four valid closed candles	INSUFFICIENT_HISTORY	May return compact candle evidence; no sequence fingerprint.
Candle missing closure proof	Validation failure	Candle excluded/rejected; never silently marked closed by the study module.
Contradictory/non-finite geometry	Validation failure	Candle excluded or entire direct call rejected.
Mixed coordinate spaces	Validation failure	Fingerprint refused.
Retracement candles lack stable closed identities or a later pivot confirmation	NO_PROVEN_COMPLETED_SWINGS	The chart study continues, but no retracement/object observation is invented or persisted.
Source timestamp jumps more than one timeframe or a changed source ID has no exact rollover chain	SOURCE_BAR_GAP_UNPROVEN	Resolver sequence remains unchanged; no purported one-candle outcome or retracement Pair DNA row matures.
Retracement object lacks explicit value bounds	NO_COMPARABLE_OBJECTS when none remain	Normalized screen geometry is not guessed into price or price-proxy values.
Retracement swing/object value spaces differ	Validation failure, then DEGRADED at tracker boundary	Cross-scale confluence is rejected; permission remains independent and fail-safe.
Stable ledger identity absent/synthetic	SKIPPED_UNSTABLE_IDENTITY at direct ledger boundary	Exact ledger remains unchanged; the service itself writes only a proven close key.
SQLite WAL unavailable, schema incompatible, lock timeout, or ledger capacity exceeded	DEGRADED at tracker boundary	The ledger transaction rolls back; capture, dashboard, and independent permission continue fail-safe.
Positional tracker ID offered as stable history identity	SKIPPED_UNSTABLE_IDENTITY or audited unstable-candle skip	Display analysis may continue; no ledger or lifelong Pair DNA mutation derives from the position.
Pair DNA receives a marker outside its locked order domain	RECORDED envelope with skipped_order_domain_conflicts incremented	Timestamp and resolver-event orders are never compared or mixed.
Pending/current resolver sequence is not exact N -> N+1	SKIPPED_UNPROVEN_ONE_STEP_HORIZON	Replay, gap, and out-of-order observations cannot mature an outcome.
Prior close not re-observed on current frame axis	SKIPPED_UNPROVEN_COORDINATE_CONTINUITY	No outcome label, historical enrichment, or Pair DNA outcome update.
No historical match above threshold	NO_MATCHES	Regression/candle study can still describe direction; similarity adds no vote.
Fewer than three labeled matches	INSUFFICIENT_OUTCOME_SUPPORT	No historical continuation vote.
Similar outcomes too close	MIXED_EVIDENCE	No forced historical side.
Pair DNA duplicate	DUPLICATE_IGNORED	Cumulative counts unchanged.
Possible Bloom duplicate	POSSIBLE_DUPLICATE_IGNORED	Counts unchanged to protect lifelong integrity.
Pair DNA candle timestamp does not advance, or segment boundary is unstable/incomplete	RECORDED envelope with audited skip counts	No duplicate candle, duration, segment, or transition aggregate is added.
Pair DNA segmented Bloom reaches 10,240 sequence identities	Capacity validation failure	Store fails closed and requests sharding; no lifelong profile is silently evicted.
Retracement study repeats through an overlapping window/restart	Existing empirical bucket remains unchanged	Stable confluence ID is rejected by the dedicated dedupe Bloom.
Retracement empirical partition capacity is reached	Capacity validation failure	Existing lifelong buckets remain intact; explicit sharding or a bounded limit increase is required.
More than 2,048 association keys	Profile remains valid	Established associations remain; unseen keys increment association_overflow_count.

Condition	Public study state	System behavior
Object graph exceeds live caps	READY_TRUNCATED	Deterministic highest-priority proven edges remain and every omission is counted.
Object graph has unsafe/conflicting geometry or identity	Validation failure, then DEGRADED at tracker boundary	No guessed anchor/overlap is published; independent capture and permission continue.
Motif, duration, or path input contains a forming candle, mixed coordinate/order scope, or an unproven gap	Validation failure	No hierarchical node, duration curve, or reconstructed path is published from discontinuous evidence.
Motif/survival/path configured capacity is exceeded	Validation failure	The service fails closed instead of truncating hidden evidence or expanding resource use.
Shadow feature promotion gate fails	Feature remains SHADOW	It is absent from the promoted-only public snapshot; failed checks remain in the versioned audit trail.
Drift evidence is insignificant or still warming	STABLE or WARMING	Current regime partition remains unchanged and no directional interpretation is added.
Drift partition capacity is reached	DRIFT_PARTITION_CAPACITY_REACHED	The existing partition history remains intact; no unbounded partition is created.
Cross-pair timestamps, normalized space, or order domain differ	Validation failure	No edge is calculated. A non-significant direction is suppressed, and every published Granger-style/MI edge remains explicitly non-causal.
Claim certificate or supplied proof material is altered	INVALID	The study claim is suppressed; certificate validity can never promote trade authority.
Pending outcome journal exceeds 64 pair/timeframe rows	Current study remains STUDIED	Oldest pending ordinal is evicted; Pair DNA, historical sequences, and exact candle rows remain.
Reacquired history event lacks an exactly matching study key and sequence	Event remains valid without a study snapshot	The newest live study is not backfilled onto the older candle.
Public app receives non-study authority flags or non-allowlisted nested fields	Study or fields omitted	No generic depth expansion exposes geometry, fingerprints, persistence, or model input.
JSON lock timeout, invalid JSON, 16 MiB overflow, or Pair DNA/historical capacity error	DEGRADED at tracker boundary	Log exception; capture, dashboard, and permission remain operational and fail-safe.
Process restart	Three JSON stores and the SQLite ledger reload; process caches reset	Exact candles, fingerprints, profiles, and bounded pending one-step maturation survive.

15. Testing and operating proof

15.1 Implemented focused tests

Test	Evidence covered
test_candle_intelligence_service_v3.py	Exact wick/body computation, rejection/acceptance, tracker proof requirement, pixel adaptation, malformed/forming failure.
test_candle_micro_geometry_v3.py	Tracker behavior tokens retain measured candle geometry and wick micro-events.
test_candle_ledger_v3.py	Rolling-window upserts, stable identity, pair/timeframe isolation, exact proxy/pixel evidence, restart/WAL, cross-process safety, and atomic capacity rollback.
test_behavioral_sequence_service_v3.py	Swing/rest segmentation, transitions, duration, major/inner trend, insufficient history.
test_pair_dna_store_service_v3.py	Monotonic unique-candle high-water, timestamp/resolver-event order-domain locking and conflict skips, stable completed-segment dedupe, retracement partitions and replay protection, Bloom probability/capacity, legacy migration, corruption and concurrent writes.
test_object_relationship_graph_v3.py	Proven anchors/overlap, confirmed-pivot retracement math in both directions, 70.5%/experimental-71.8% object confluence, coordinate rejection, deterministic caps/truncation, unsafe geometry failure, and stripped trade instructions.
test_historical_similarity_service_v3.py	Fingerprint validation, scoring, bounded storage/graph, supported continuation, and correlation statistics.
test_market_study_service_v3.py	Idempotent close-key caching, exact candle-ledger integration, exact +1 horizon enforcement, delayed retracement maturation, restart maturation, continuous research-key publication, deterministic ontology/drift rebuild, exact cross-pair synchronization/abstention across restart, current-frame pixel-axis proof/skip, bounded pending journal, and no execution authority.
test_market_study_tracker_bridge_v3.py	Live resolver-to-study promotion: positional IDs stay unstable, stable history accumulates across source and screenshot rollovers, production objects require stable candle anchors, object boxes publish the exact candle-compatible pixel-value axis, spoofed caller stability cannot promote an object, and an arbitrary reacquired pixel window cannot false-mature.
test_market_study_operator_integration_v3.py	Live/operator explicit nested allowlist, bounded retracement summary, experimental-level disclosure, privacy boundary, and fixed V3 schema integration.
Focused test_window_tracker_service.py cases	Real source keys remain distinct and pixel bounds/points normalize against exact image dimensions before the study graph.
test_motif_lattice_v3.py	Deterministic four-level motifs, contiguous order proof, bounded node publication, censored time-to-event curves, and anchor-known path reconstruction with MFE/MAE/state-time evidence.
test_advanced_research_studies_v3.py	Shadow feature gates and rollback, deterministic drift partitions, exact-timestamp normalized cross-pair association, non-causal contracts, graph bounds, and proof-certificate tamper rejection.
test_dashboard_static_contract.py	Required V3 study and label-mode DOM/static contracts.
test_dashboard_collision_playwright.py	Exhaustive labels, retired forecast families, market-study schema use, and regression-history behavior in a browser.
Existing operator/live-state tests	Fixed public schemas, compact projection, privacy, history, freshness, and permission separation.

15.2 Required release validation

Run focused tests first, then the complete repository suite. The final acceptance pass should prove:

- 1 same close key returns a byte-equivalent logical study and does not add a second exact candle or increment Pair DNA;
- 2 a new close matures only the exact preceding resolver event ($N \rightarrow N+1$) when that prior close is uniquely re-observed on the current frame axis; replay, gap, out-of-order, autoscale/pan, and missing continuity must skip;
- 3 process restart reloads all three JSON stores and the SQLite WAL ledger without schema drift;
- 4 concurrent JSON and SQLite writers do not lose increments or publish partial state;

- 5 overlapping windows count stable candles and completed segment transitions only once; Pair DNA locks one timestamp-or-resolver-event order domain and audits every cross-domain conflict;
- 6 segmented Bloom design remains below its declared union false-positive ceiling and fails closed at 10,240 identities;
- 7 corrupt, oversized, non-finite, incompatible-schema, and capacity-bound stores fail closed;
- 8 real object keys remain distinct display identities; exact image dimensions normalize pixel bounds/points; anonymous and keyed-but-unproven objects stay OBSERVATION_ONLY; graph anchors/overlaps require explicit proof and all edges remain bounded, study-only, and non-causal;
- 9 live-state and operator APIs expose the same current study identity through the explicit app allowlist without raw geometry/fingerprint leakage;
- 10 one real broker rollover creates one and only one new sequence ID and exact ledger close;
- 11 the right-most forming candle never appears in durable candle intelligence;
- 12 reacquired history rows receive only an exactly matching key/sequence study snapshot;
- 13 Show all + Labels on leaves no collision-hidden or policy-hidden label visually suppressed;
- 14 retired forecast families remain absent even under Show all;
- 15 the dashboard shows major trend, inner trend, regression, and permission as separate concepts;
- 16 continuous history describes directional/rest behavior rather than defaulting every row to WAIT;
- 17 canonical .venv-live launch keeps one API/tracker topology and writes study data outside the launcher-cleaned runtime tree.
- 18 a multi-rollover current-frame binding chain produces a proven completed swing and production object confluence, while spoofed caller stability flags cannot promote an object.

Useful focused command:

POWERSHELL DIAGRAM / CONTRACT

```
$env:PYTHONPATH='Backend/src;Backend'
& '.\venv-dev\Scripts\python.exe' -m pytest `
  Backend/tests/test_candle_intelligence_service_v3.py `
  Backend/tests/test_candle_micro_geometry_v3.py `
  Backend/tests/test_candle_ledger_v3.py `
  Backend/tests/test_behavioral_sequence_service_v3.py `
  Backend/tests/test_scene_forecast_contributor_v3.py `
  Backend/tests/test_pair_dna_store_service_v3.py `
  Backend/tests/test_object_relationship_graph_v3.py `
  Backend/tests/test_historical_similarity_service_v3.py `
  Backend/tests/test_market_study_service_v3.py `
  Backend/tests/test_market_study_tracker_bridge_v3.py `
  Backend/tests/test_market_study_operator_integration_v3.py `
  Backend/tests/test_motif_lattice_v3.py `
  Backend/tests/test_advanced_research_studies_v3.py `
  Backend/tests/test_dashboard_static_contract.py `
-q
```

The canonical live launch is Backend/launch/launch_phoenixguard_live_ready.ps1. A healthy launch must expose the same closed_candle_key and sequence_id through tracker, compact live state, operator workspace, and dashboard.

15.3 Release evidence collected on 2026-07-24

Gate	Result
Repository regression collection	The 173-file regression baseline was executed in bounded deterministic shards: 2,191 passed and 35 skipped. The final inventory is 174 files because the resolver-to-study bridge regression file was added during the closing audit.
Final-code intelligence delta	111 focused tests passed after the closing audit across resolver identity, exact N -> N+1 maturation, order-domain isolation, candle micro-intelligence, behavior, exact ledger, Pair DNA, historical similarity, object graph, operator projection, UI contracts, and tracker geometry. One stale assertion that still required the retired forecast-family mapping was corrected to assert its removal and rerun green.
Static quality	Ruff passed for every changed Python file; strict Pyright on every changed backend source reported 0 errors and 0 warnings; git diff --check passed.
V3 identity	verify_v3_integrity.py reported Overall: PASS and retained V3 as the only active version.
Live API smoke	The .venv-live FastAPI runtime returned HTTP 200 for the V3 dashboard, operator workspace, compact live state, and tracker session endpoints.
Live browser proof	Chromium rendered the live dashboard with simple-view labels-show-all labels-on; 50 of 50 present labels were visible, including 44 labels carrying the policy-shadow class.
Retired UI proof	Live DOM contained no Forecast navigation, LSTM/scene forecast controls, Run forecast action, Show future path action, or Path A/Path B lane rows.
Study/history proof	Live DOM exposed Major trend, Inner trend, Regression study, separate CLOSED entry permission, and 29 regression-history rows.
Blueprint artifacts	Both architecture PDFs rendered successfully and passed metadata, text extraction, empty-page checks, and sampled-page visual inspection.

The canonical cold launcher is intentionally destructive to runtime/live; it was not used for this smoke because preserved operator artifacts were already present. Full-launch topology remains covered by the V3 launcher/integrity and window-tracker tests, while the smoke used the production .venv-live application and real HTTP/browser surfaces without deleting those artifacts.

15.4 Retracement-confluence delta evidence collected on 2026-07-25

Gate	Result
Completed post-fix matrix	328 tests passed: 122 retracement/graph/Pair-DNA/service/tracker/operator/static-contract tests; 75 adjacent candle/ledger/behavior/similarity/history tests; 88 scene-forecast/public-operator tests; 3 changed Playwright cases; 9 V3 integrity tests; 2 hardened object-identity cases; and 29 tracker payload/model-council tests.
Identity hardening	Arbitrary object IDs and zone keys remain OBSERVATION_ONLY unless a stable proof exists. The two directly changed cases passed. A broader 255-test tracker file produced about 181 passing indicators with no failure marker before the 900-second harness limit, so it is not counted as a completed pass.
Static quality	Ruff, compileall, git diff --check, and verify_v3_integrity.py passed. Strict Pyright reported 0 errors and 0 warnings across every changed production source and focused test.
Browser contracts	The three changed retracement evidence states passed after the final fix. The broader collision suite had already completed at 55 passed and 35 skipped before the final backend-only compaction and identity changes.
Data-preserving live launch	The direct .venv-live launcher preserved the existing pocket-live-8788 history, captured a fresh desktop frame, and passed the one-stack process-topology certificate on port 8793 with the MT4 bridge required. The destructive cold launcher was not used.
Runtime lineage	runtime_trace_v3.py reported Alignment: PASS, a current study packet, 7/7 models, and no published execution packet.
Operator DTO	On live frame 236 the operator exposed USD/CAD OTC, M5, and STUDIED; both OTE_70_5 and CUSTOM_71_8 were present. Study execution authority and entry permission were both false.
Honest zero-evidence state	The current frame reported NO_PROVEN_COMPLETED_SWINGS, zero completed graph/Pair-DNA support, and no empirical partitions. This is the required result when the current closed-candle proof cannot support a completed swing; deterministic tests separately prove populated confluence and maturation paths.

15.5 Operational telemetry to retain

- study status and reason;
- symbol/timeframe/closed key/sequence ID;
- studied candle count and coordinate space;
- exact ledger insert/update/change counts, unique candle count, and total re-observation count;
- similarity match count and labeled support;
- Pair DNA observation/candle counts, timestamp/segment-boundary skip counters, Bloom insertions and remaining identity capacity, and association overflow count;
- outcome maturation status and coordinate-continuity proof;
- pending journal entry count/size and oldest-ordinal eviction;
- object relationship node/edge/relation counts and truncation counts;
- store read/write latency, JSON/SQLite lock timeout, file size, WAL mode, and ledger capacity;
- cache hit versus new-close computation;
- degraded exceptions without raw broker imagery or secrets.

16. Advanced continuous V3 research services

The following boundary is deliberate. PhoenixGuard uses deterministic, explainable, bounded algorithms so every study claim can be traced. The services below are implemented V3 backend research primitives integrated into MarketStudyServiceV3. The app and operator boundaries publish only their bounded, privacy-allowlisted summaries; raw histories, feature windows, persistence internals, and private model contributions remain private. None can alter the independent permission contract.

16.1 Established continuous study baseline

- exact candle body/wick micro geometry in price, normalized proxy, or pixel proxy space;
- candle taxonomy, personality, rejection/acceptance, and prior-candle relations;
- automatic closed-candle-only continuous study with idempotent repeated frames;
- SQLite WAL exact candle ledger with one authoritative stable close per study and exact unique/ re-observation counts;
- major/inner regression and explicit swing/rest segmentation;
- persistent bounded Pair DNA aggregates by pair and timeframe with monotonic candle high-water, stable segment-boundary dedupe, and segmented replay protection;
- object/candle association counts with non-causal labeling;
- bounded current-study object relationship graph with explicit anchors, observed-with, co-occurrence, and proven normalized-overlap edges;
- explainable 60-value normalized sequence fingerprint;
- deterministic pair-scoped nearest-sequence search and bounded similarity graph;
- one-candle delayed outcome maturation and minimum-support historical continuation;
- one BUY/SELL/HOLD directional study with a hard study/execution boundary;
- compact live-state, explicit app allowlist, exact continuous-history attribution, operator, and chart-led frontend contracts;
- retired public forecast families plus exhaustive `Show all + Labels` on label rendering.

16.2 Seven implemented research-grade capabilities

Capability	Exact behavior	Bounds and honesty contract
Hierarchical multi-resolution motif lattice	<code>build_hierarchical_motif_lattice_v3(...)</code> composes level 0 single-candle micro-events, level 1 3-5 candle atoms, level 2 7-12 candle compounds, and level 3 full swing/rest regimes. Parent nodes retain bounded child composition and normalized features.	Depth is at most four, nodes at most 2,048 per level, children at most 64 per node, and input history at most 512 closed candles. Motifs describe history; resemblance is not a future outcome.
Historical time-to-event evidence	<code>build_time_to_event_survival_evidence_v3(...)</code> measures next swing, direction change, and rest end. It publishes Kaplan-Meier-style curves, at-risk/event/censor counts, restricted mean duration, median when observed, and Greenwood log-log intervals. Object-conditioned edges are built only from durable matured Pair DNA sequences; without that history the service explicitly abstains.	At most 32 histories, 49,152 derived observations, and 256 closed candles of horizon. Overlapping origins may be dependent. The curves are descriptive, non-causal, and not future deadlines.
Adaptive feature ontology	<code>AdaptiveFeatureOntologyV3</code> creates pair/timeframe-scoped shadow proposals with bounded JSON definitions and closed-candle evidence. The live service calculates train/holdout support, stability, effect size and multiplicity-adjusted evidence internally before promotion. Promotion and rollback append immutable audit revisions, persist restart-safely, and the public snapshot excludes raw shadow definitions.	At most 2,048 features and 32 revisions by default. Passing the gate means eligible for public study only; it neither establishes causation nor authorizes prediction or execution.
Exact normalized historical path reconstruction	<code>reconstruct_normalized_historical_path_v3(...)</code> fixes scale from candles known at the anchor, reconstructs later close/high/low paths in median-range units, and records MFE, MAE, final displacement, efficiency, transitions, and time in state. The service also publishes a bounded trajectory library linked to historical motif/similarity evidence.	Every path is independently fail-closed and clamped to at most 256 candles, so one invalid path cannot suppress motif or survival results. Future geometry cannot alter normalization. Direction is a reference orientation, not a trade instruction.
Multi-pair association graph	<code>analyze_cross_pair_lead_lag_v3(...)</code> and <code>build_cross_pair_association_graph_v3(...)</code> require distinct pairs, exact shared contiguous closed timestamps, one compatible normalized coordinate space, and one order domain. <code>CrossPairStudyCoordinatorV3</code> atomically retains bounded normalized-return tails as pair updates arrive, aligns only genuine exact timestamps, then invokes that engine. They compare an autoregressive baseline with a lagged-source augmentation, measure mutual information, and test the selected lag against deterministic circular shifts.	The pure engine permits at most 1,024 samples, lag 12, eight graph pairs, and 64 published edges; the live coordinator defaults to 256 samples, lag six, eight pairs, 32 edges, and 63 null shifts so an adjusted 5% result is mathematically reachable. Without compatible support it abstains. Granger-style variance reduction and mutual information are non-causal associations only.
Online concept-drift partitioning	<code>OnlineConceptDriftDetectorV3</code> compares fixed adjacent closed-candle feature windows with a two-sample KS statistic, Bonferroni-adjusted DKW threshold, and standardized mean-shift floor. Digest-validated snapshots replay idempotently; a significant change closes the prior partition and persists a deterministic append-stable regime ID into Pair DNA.	Window size is at most 256, feature count at most 64, and partition history is bounded. A partition boundary describes distribution change only and never market direction.
Machine-checkable claim proofs	<code>issue_study_claim_certificate_v3(...)</code> hashes the claim, bounded inputs, derivation, ordered closed-candle evidence, coordinate space, and order domain into a deterministic certificate. A non-circular <code>PG_PUBLIC_STUDY_CANONICAL_V3</code> projection keeps the bound hash stable after proof references are attached, and cross-pair certificates bind every tested peer digest.	At most 512 evidence candles, bounded depth/collection/document sizes, and allowlisted material public claim types. Verification rejects envelope, evidence, peer-manifest or final-study tampering. A valid certificate proves supplied-evidence integrity only; it does not authenticate the market source, prove causation or accuracy, or grant execution permission.

All seven services require one pair/timeframe and one coordinate/order domain unless the cross-pair service explicitly validates compatible normalized streams. All return `study_only: true`, `causal: false`, and `execution_authority: false` or an equivalent hard contract. Capacity overruns fail closed; no service silently expands its bound or invents missing continuity.

16.3 Continuous pipeline integration and promotion boundary

- 1 MarketStudyServiceV3 supplies only identity-proven closed histories. A forming candle, gapped order, mixed coordinate space, or positional identity fails validation.
- 2 Every successful continuous study carries the bounded keys `motif_lattice`, `survival_network`, `path_reconstruction`, `adaptive_feature_ontology`, `concept_drift`, `regime_partition`, `cross_pair_association`, and `claim_proofs`. Pending studies carry the same families with explicit insufficient-evidence states rather than invented findings.
- 3 Ontology audit state persists per pair/timeframe after internally calculated train/holdout gates. Drift snapshots and append-stable regime partitions persist inside Pair DNA and replay idempotently. Only promoted ontology definitions enter the public snapshot; shadow totals remain bounded audit metadata.
- 4 Cross-pair normalized returns use their own versioned atomic coordinator store. A lone pair, timestamp mismatch, incompatible timeframe, or insufficient support must abstain with an explicit status rather than manufacture a peer, shared timestamp, or edge.
- 5 The dedicated app/operator allowlist publishes bounded research summaries only after removing raw identities, raw geometry, full feature windows, dedupe state, and persistence internals.
- 6 The ontology's temporal-safety gate is an eligibility screen. Documentation and UI must never abbreviate it to proof that a feature causes an outcome.
- 7 Cross-pair edges must always travel with support, lag, coordinate space, order domain, significance method, and a visible non-causal statement.
- 8 Proof certificates travel with the claim they bind and the declared derivation version. A digest mismatch suppresses the claim; a valid digest still cannot elevate authority.
- 9 Public evidence remains separate from current entry permission, execution-packet validation, and downstream bridge controls.

16.4 Further research-informed shadow challengers

These remain future challengers, not claims about the current implementation:

- A [Matrix Profile](#) challenger can discover motifs and discords in normalized candle-feature streams while preserving an inspectable nearest subsequence.
- A [learnable multivariate Matrix Profile](#) challenger can test cross-dimensional candle/object motifs and learned inter-feature structure. It remains an offline, leakage-controlled shadow until it beats the explainable same-pair baseline without weakening boundedness or auditability.
- Constrained multivariate dynamic-time-warping motif search can test whether elastic timing improves swing/rest matches; see the primary [DTW motif-search research](#). It must be bounded and compared against the current fixed resampling baseline.
- [MiniROCKET](#) or its [ROCKET](#) parent can provide a fast supervised sequence classifier challenger after leakage-safe offline training.
- [TS2Vec](#) can provide a self-supervised multiscale embedding challenger, but only a versioned, frozen encoder trained without future/test leakage may enter a live shadow lane.
- [Bayesian online change-point detection](#) can challenge the deterministic swing/rest boundary when it provides a causal online posterior rather than a hindsight segmentation.
- [Adaptive conformal prediction for time series](#) and [adaptive conformal inference under distribution shift](#) can challenge uncertainty calibration under regime drift. Coverage must be measured sequentially; conformal output still cannot grant permission.

Every challenger must publish beside the baseline with its model/data version, training cutoff, pair/timeframe scope, latency, abstention behavior, calibration, and exact evaluation split. It is promoted only when it improves out-of-sample decision support without weakening closed-candle temporal integrity, explainability, runtime bounds, or the independent execution gate.

17. Joint Path-Clock Liquidity Field V3

17.1 Timing question and admission boundary

Directional correctness is not enough for a fixed-duration OTC contract. The timing lane must study the joint event:

TEXT DIAGRAM / CONTRACT

```
the selected stop remains unbroken
AND the studied directional move reaches its target
AND that event occurs inside the declared contract clock
```

PG_PATH_CLOCK_LIQUIDITY_V3 therefore conditions historical evidence on three simultaneous axes:

- 1 Path: signed displacement, high excursion, low excursion, MAE, and MFE in median-range units fixed at the anchor.
- 2 Clock: elapsed seconds and remaining seconds under one explicit fixed-duration contract.
- 3 Liquidity state: a bounded five-axis vector containing wick entropy, repeated-area touches, late-sweep motif distance, body/wick asymmetry, and object co-presence density.

The canonical duration policy is PG_TIMING_DURATION_POLICY_V3:

- a new timing anchor is admitted only when the declared duration is at least 900 seconds;
- the bounded maximum is 7,200 seconds;
- 899.999 seconds remains excluded and exactly 900 seconds is eligible;
- an already admitted anchor continues to be observed when its remaining clock falls below 900 seconds, because the final-clock sweep is material evidence;
- a new entry is not eligible from a late anchor with less than 900 seconds remaining;
- a move that reaches its target before 900 elapsed seconds is retained in the audit trail but is excluded from eligible timing, calibration, and promotion statistics;
- direction evidence from a shorter horizon may still describe the chart, but it is never timing authority for a sub-15-minute trade.

17.2 Closed-candle anchor and trajectory schema

Every anchor identity binds all of the following fields:

TEXT DIAGRAM / CONTRACT

```
schema + feature version
pair + timeframe
coordinate space + order domain
stable anchor close key + monotonic close order
anchor close timestamp + source cadence
studied direction + explicit contract duration
anchor-known median-range scale + liquidity vector
```

Every later observation is uniquely identified by (anchor_id, observation_order). An identical retry is an idempotent no-op; a conflicting payload under the same identity fails closed. One trajectory contains only observed closed-candle timestamps. A 30-second grid is permitted only when the source cadence or an exact sub-candle source proves those timestamps. PhoenixGuard never interpolates M5 OHLC into invented 30-second timing evidence.

The close clock is a separate proof from candle identity. source_bar_id may prove which bar was observed while closed_candle_epoch, or a source open time plus the proven timeframe, proves when it closed. Both are retained even when they appear on the same source row. A generic frame capture time, publish time, frame number, X position, or initial screenshot baseline is never relabeled as a candle close.

For screenshot-only streaming, V3 can create PG_PROVEN_CLOSED_CANDLE_TIME_V3 only after the resolver proves exactly one forming-to-closed transition and two capture observations bracket exactly one timeframe boundary. The certificate binds pair, timeframe, stable close key, monotonic close sequence, exact bound row, source cadence, boundary epoch, observation latency, transition count, and proof source. Multi-candle reacquisition, an unbracketed boundary, a clock gap, conflicting source times, or a pair/timeframe change yields no certificate. The bounded event-time ledger is persisted with the closed-candle identity state and reattaches time only to the exact stable bindings reobserved on the current geometry axis.

The normalized point contract is:

TEXT DIAGRAM / CONTRACT

```
elapsed_seconds
path_mru
high_mru
low_mru
```

where MRU means median-range unit frozen from information available at the anchor. The completed trajectory records stop-before-target order, target time, MAE, MFE, final directional displacement, path efficiency, and clock-conditioned adverse/favorable excursions. When OHLC cannot prove the intrabar order of a stop and target touched within one candle, the result is UNKNOWN_FAIL_CLOSED; it is never guessed from candle color.

17.3 Causal maturation and discontinuities

An anchor matures only when an observed closed candle proves elapsed time equal to its declared duration. The side store censors, instead of completing, an anchor when any of these occur:

- pair or timeframe changes;
- coordinate space or order domain changes;
- stable close order is non-monotonic or has an unproved gap;
- the exact expiry boundary was skipped;
- the anchor-known scale is missing or invalid;
- bounded anchor or trajectory capacity is reached.

Forming-candle wick/body measurements may appear in transient watch telemetry, but they never enter the durable field, Pair DNA, replay score, calibration, or permission. Restart restoration fully revalidates every identity, point, bound, digest, and safety flag before the field can be queried.

17.4 Historical field and query contract

For a live closed-candle state, the engine finds bounded same-pair neighbors compatible with the direction, duration, liquidity vector, coordinate space, order domain, and causal cutoff. For each declared stop/target scenario it publishes a compact estimate containing:

- support count and effective neighbor distance;
- stop-before-target survival probability;
- probability that the worst historical pullback still lay ahead at this clock position;
- target-arrival and adverse-excursion distributions;
- eligible and excluded-early outcome counts;
- explicit remaining clock and new-entry eligibility;
- an abstention status when support or continuity is insufficient.

Clock-field rows are keyed by both the declared contract duration and the aligned elapsed/remaining clock. Two trajectories at the same elapsed second but with different expiry clocks are never merged into one timing population. The published probability that the worst drawdown still lies ahead counts only histories whose global maximum adverse excursion occurs after the aligned current point; any later adverse movement is not sufficient by itself.

The wording “worst pullback” is used in the operator view. It describes the adverse excursion of the studied price path, not account drawdown and not a guaranteed future sweep.

17.5 Freeze, replay, calibration, and promotion

Every proven close freezes its field state before later candles are read. The freeze binds the exact historical-library revision and digest used for that estimate, so adding later trajectories cannot rewrite an earlier claim. Replays then score four independent axes:

Axis	Required measurement
Direction	Did the final realized path agree with the frozen studied direction?
Timing	Did the move occur inside the frozen survival window?
Sweep survival	Did each declared stop remain alive until the move occurred?
Calibration	Did predicted survival frequencies match realized frequencies?

A candidate timing policy is promoted only after the minimum replay support is present and it improves all four axes against the registered baseline. One strong direction score cannot hide poor timing, worse sweep survival, or miscalibration. Promotion is versioned, reversible, bounded, and pair/timeframe scoped. Until the gate passes, the field remains useful historical study but cannot support or veto an otherwise valid entry, except for the hard duration contract itself.

Every exact completed horizon enters the replay cohort, including paths that never reached the frozen target and paths that finished flat. A no-target horizon is right-censored at its exact expiry with `observed_move_occurred=false`; it is a timing miss, every target-before-stop outcome is false, and its survival probability contributes a negative calibration observation. It cannot disappear from support or count as a success. Candidate and baseline scores must prove the same ordered closed-candle cohort, and each sweep outcome binds the same stop distance and move size on both sides. A cohort mismatch or scenario mismatch fails promotion rather than allowing stop widening to masquerade as model improvement.

17.6 Persistence and Pair DNA boundary

Raw normalized paths and frozen replay states live in a dedicated atomic side store under `path_clock_liquidity_v3`. They are never embedded into the shared Pair DNA JSON. Default bounds are:

Resource	Default hard bound
Historical trajectories	256
Normalized points per trajectory	241
Frozen closed-candle states	512
Nearest neighbors per query	64
Active pending anchors	bounded by the side-store contract
Contract horizon	900-7,200 seconds

Pair DNA receives only a compact partition summary: counts, duration distribution, direction distribution, calibration support, last update identity, and a digest. Side-store writes are locked, validated, size-bounded, written to a same-directory temporary file, flushed, fsynced, and atomically replaced. A corrupted or oversized snapshot is rejected rather than partially loaded.

17.7 Public timing and the independent permission contract

The public allowlist exposes only the current bounded timing read: lineage, side, duration, remaining clock, support, calibrated probabilities, promotion status, and safety flags. Raw paths, neighbors, liquidity vectors, detector state, geometry, hashes, and persistence internals are private.

JPCLF is an asymmetric gate:

Existing entry permission	Mature JPCLF timing	Q3 result
Closed	Eligible now	Timing may be ready, but permission remains closed.
Open	Wait-until state	Do not enter yet; show the earliest eligible clock.
Open	Sweep-risk or worst-pullback-ahead veto	Stand aside; name the timing risk.
Open	Eligible now	Existing entry may remain open, subject to every other V3 gate.
Either	Missing, immature, stale, or mismatched	Ignore JPCLF; retain the pre-existing permission result.

The public contract keeps four separate booleans:

TEXT DIAGRAM / CONTRACT

```

permission_allowed
timing_supports_entry
timing_veto
entry_permission_authorized

```

timing_supports_entry can never turn a false permission_allowed into authorization. Timing can delay or veto; it cannot create a BUY or SELL. Q1 remains the history answer. Q2 keeps direction authoritative and may append one plain timing sentence only after the field is mature. Q3 remains the single best action now. The streaming refresh applies the same gate, so a heartbeat cannot erase a timing veto or revive stale timing from another pair, frame, or close.

17.8 Failure modes and test matrix

Failure or test	Required outcome
899-second candidate	Excluded before ranking, support, confidence, or persistence
Exact 900-second candidate	Eligible for a new anchor
Admitted anchor reaches 120 seconds remaining	Observation continues; new-entry eligibility is false
Same anchor/observation replayed	Exact duplicate is a no-op; conflicting duplicate fails
Source bar id plus source timestamp	Preserve and validate both independent proofs
Initial screenshot baseline	No candle-close timestamp is invented
One resolver close bracketed across one timeframe boundary	Emit one key/sequence/row-bound close-time certificate
Capture/publish time without a close transition	Never becomes candle time
Locked Chromium surface remains byte-identical beyond the bounded threshold	Treat it as capture-health evidence, not market rest; rate-limit an identity-verified visible recovery with no clicks or keys, and admit one recovery keyframe only after the observer proves non-duplicate pixels
Multi-candle reacquisition or missed boundary	Censor; do not backfill or interpolate timestamps
Greatest timestamp belongs to an unbound row	Reject it; current event is selected by proof key and sequence
Forming candle changes	No durable field, replay, or Pair DNA mutation
Pair/timeframe/domain/coordinate change	Pending anchors are censored and old timing is removed from public state
Process restart	Digest-validated field and pending anchors restore exactly once
Target and stop touched in one unsequenced candle	Intrabar order is unknown and fails closed
Public projection	No raw trajectory, neighbor vector, geometry, or private digest leaks
Timing says enter but permission is closed	Q3 stays closed
Permission is open but mature timing vetoes	Q3 becomes do-not-enter without mutating permission truth
Stream heartbeat	The same timing gate is reapplied with current lineage and freshness
Capacity reached	Old bounded evidence is evicted/censored according to policy; memory cannot grow

18. Definition of done for this V3 lane

The V3 market-study lane is complete only when all of these statements are true at the same time:

- one proven close creates at most one durable study event;
- every persisted candle has explicit closure proof, a stable pair/timeframe close identity, and valid exact geometry; rolling windows upsert rather than duplicate it;
- screenshot positional IDs remain display-only; resolver identity persists only through the bounded current-frame stable_visible_candle_bindings contract under PG_CLOSED_CANDLE_IDENTITY_STATE_V3, a stable event key, and monotonic event sequence;
- a byte-identical off-screen browser cache cannot masquerade as a quiet market or advance JPCLF; the explicitly locked chart uses bounded, identity-verified visible recovery and no input synthesis;
- the operator can see major trend, inner trend, current swing/rest behavior, and the one studied directional read;

- Pair DNA counts only new monotonic candles and proven completed-segment boundaries inside one locked CLOSED_TIMESTAMP_V1 or TRACKER_EVENT_SEQUENCE_V3 order domain; historical examples and the exact candle ledger remain distinct, durable, bounded, and pair/timeframe scoped;
- an outcome matures only across an exact resolver +1 event after its prior close is uniquely re-observed on the current frame axis; shifted/reacquired windows cannot substitute for proof;
- real object keys and exact-dimension-normalized geometry remain distinct; anonymous and keyed-but-unproven identities stay OBSERVATION_ONLY, and relationships never infer anchors, causation, or permission;
- historical continuation is unavailable until causal outcomes reach minimum support;
- every association and match can be explained from stored evidence;
- motif nodes compose deterministically through all four declared levels and never exceed their published depth/node/child bounds;
- time-to-event curves retain support, at-risk events, censoring, and uncertainty and never present a historical duration distribution as a deadline;
- reconstructed paths use only anchor-known normalization and disclose MFE, MAE, efficiency, transitions, and state time as historical evidence;
- shadow features cannot enter a public study snapshot until their latest versioned gate passes, and every promotion can be rolled back without deleting audit history;
- drift creates a new deterministic regime partition only from significant closed-window evidence and never supplies directional permission;
- every cross-pair edge uses exact shared closed timestamps and compatible normalized geometry, and all Granger-style and mutual-information language remains explicitly non-causal;
- every published proof certificate validates its claim, inputs, derivation, ordered closed-candle identities, coordinate space, and order domain, while retaining zero execution authority;
- retired forecast families cannot return through Show all, and Show all plus Labels on really shows every present overlay label, even when clustered;
- session history describes what price did instead of mirroring permission WAIT;
- session history never attaches a same-frame study to a different close key/sequence;
- entry permission remains independent, fresh-frame gated, and fail-closed;
- the complete V3 suite and live topology checks pass without creating any V4 surface.